Entropy and Gibbs Free Energy

ΔS, ΔG = ΔH − TΔS and K

Name _____ Class _____ Date _____

EQUATIONS AND DATA (298.15 K)

ΔS° = ΣnS°(products) − ΣnS°(reactants) · ΔG = ΔH − TΔS (T in K; ΔS in kJ/K) · ΔG° = −RT ln K · ΔG° = −nFE°

R = 8.314 J mol^{-1} K^{-1} · F = 96 485 C/mol

S° (J mol^{-1} K^{-1}): $CH_4(g)$ 186.3 · $O_2(g)$ 205.2 · $CO_2(g)$ 213.8 · $H_2O(l)$ 70.0 · ΔHf° (kJ/mol): $CH_4(g)$ −74.6 · $CO_2(g)$ −393.5 · $H_2O(l)$ −285.8

Show units on every quantity in every step.

A. Understanding

1. (a) What is entropy? (b) State the second law of thermodynamics. (c) Why can an endothermic process, such as ice melting at 25 °C, be spontaneous?

2. Predict the sign of ΔS: (a) $H_2O\,(l) \longrightarrow H_2O\,(g)$ (b) $2\,SO_2\,(g) + O_2\,(g) \longrightarrow 2\,SO_3\,(g)$ (c) $NaCl\,(s) \longrightarrow Na^+\,(aq) + Cl^-\,(aq)$ (d) $CaCO_3\,(s) \longrightarrow CaO\,(s) + CO_2\,(g)$ (e) $N_2\,(g) + O_2\,(g) \longrightarrow 2\,NO\,(g)$

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3. For each reaction, say whether it is spontaneous at all temperatures, never, only at low T, or only at high T: (a) $\Delta H = -120$ kJ, $\Delta S = +85$ J/K (b) $\Delta H = +45$ kJ, $\Delta S = -30$ J/K (c) $\Delta H = -60$ kJ, $\Delta S = -150$ J/K (d) $\Delta H = +90$ kJ, $\Delta S = +200$ J/K

B. Calculations

4. Calculate ΔS° for $\mathrm{CH_4\,(g) + 2\,O_2\,(g) \longrightarrow CO_2\,(g) + 2\,H_2O\,(l)}$ and explain its sign.

5. Calculate ΔH° and ΔG° at 298.15 K for the same reaction.

6. For melting ice, $\Delta H = +6.01$ kJ/mol and $\Delta S = +22.0$ J mol^{-1} K^{-1}. (a) At what temperature is $\Delta G = 0$? (b) Find ΔG at -10.0 °C and at $+10.0$ °C, and explain what each result means.

7. For $\mathrm{CaCO_3 \longrightarrow CaO + CO_2}$, $\Delta H^\circ = +179.2$ kJ and $\Delta S^\circ = +160.2$ J/K. Calculate ΔG° at 1200 K. Is the decomposition spontaneous?

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8. A reaction has $\Delta G^\circ = +10.0$ kJ at 298.15 K. Calculate K, and say whether reactants or products are favoured.

C. Challenge

9. The Daniell cell has $E^\circ = 1.10$ V and $n = 2$. Calculate (a) ΔG° (b) K at 298.15 K. What does the size of K tell you?

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